

Practice 2:

Documentation and Dependency Management

Course: Software Development Oriented to Machine Learning

Overview

In this practice, you'll add documentation and dependency management to your ML project. You'll create comprehensive inline documentation and generate deployment-ready HTML documentation using `pdoc` or `sphinx`. Additionally, you'll implement proper dependency management to ensure your project can be easily reproduced on any system.

Objectives

Learn to document code professionally using docstrings and automated documentation generation. Employ dependency management for reproducible ML environments. Focus on creating documentation that helps others (and future you) understand and use your code effectively.

Requirements

The new requirements for this part of the project are:

Code Documentation: Add comprehensived inline documentation to all your modules, classes, and functions using proper docstrings. Follow established conventions like Google, NumPy, or Sphinx style (whatever looks better to you, but keep it consistent). Document function parameters, return values, and **provide usage examples** where it could be helpful.¹

HTML Documentation: Set up `pdoc` to automatically generate HTML documentation from your docstrings. `Pdoc` lets you configure the format of your docstrings, but try to make sure you have the HTML export working before documenting everything. The generated HTML documentation should be stored in a dedicated folder (usually `docs/` or `documentation/`) within your project structure. You can also use `Sphinx`, a more profesional tool, but the setup process is a bit more complicated. **READ AND CHECK** your output documentation: would you be able to navigate it like the Numpy / Sklearn / PyTorch... documentation?

Dependency Management with uv: Employ `uv` for managing project dependencies and virtual environments. If you prefer a different dependency management system (Poetry, Pipenv, conda,

¹Feel free to use an LLM to *help* you with the documentation. I recommend using a command line tool like `aider` or `crush` so that you can also try that approach, always revieweing all of the comments that it adds.

etc.), you may use it instead, but always document the choice at your `README.md`. Ensure that running `uv sync` (or equivalent) on a fresh system is sufficient to set up the entire environment and run the code. Remember that the default behavior of `uv` probably allows too much freedom in version specification. Pin major versions but allow minor updates unless you have specific compatibility requirements.

Evaluation Criteria

This practice contributes to your overall project assessment but won't be graded individually. Not handing it in will deduct from the final grade.